

session 22 and 23 Geometry Solid shapes Digital sat



$$AB = 16 \text{ cm}$$

$$h = 14 \text{ cm}$$

$$r = ?$$

1. $r = ?$
- | | |
|---------|---------|
| a. 11.7 | c. 21.2 |
| b. 1.6 | d. 7.7 |

2. A right circular cone has a volume of $2,700\pi$ cubic centimeters and the area of its base is 225π square centimeters. What is the slant height, in centimeters, of this cone?

- | | |
|-------|-------|
| a. 12 | c. 36 |
| b. 15 | d. 39 |

3. The diameter of the base of a cone is 54 units, and its height is also 54 units. What is the cone's volume in cubic units?
- | | |
|----------------|-----------------|
| a. $13,122\pi$ | c. $52,488\pi$ |
| b. $39,366\pi$ | d. $157,464\pi$ |

A right circular cone has a volume of $2,700\pi$ cubic centimeters and a base area of 225π square centimeters.

What is the slant height of the cone, in centimeters?

- _____ 4.
- | | |
|-------|-------|
| a. 12 | c. 36 |
| b. 15 | d. 39 |

A sample of oak has a density of 807 kilograms per cubic meter. The sample is in the shape of a cube, where each edge has a length of 0.90 meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

- _____ 5.
- | | |
|--------|----------|
| a. 588 | c. 897 |
| b. 726 | d. 1,107 |

A cube has an edge length of 41 inches. What is the volume, in cubic inches, of the cube?

- _____ 6.
- | | |
|----------|-----------|
| a. 164 | c. 10,086 |
| b. 1,681 | d. 68,921 |

- _____ 7. A cube has an edge of 41 in. What is the volume of the cube?
- | | |
|---------|----------|
| a. 164 | c. 10086 |
| b. 1681 | d. 68921 |

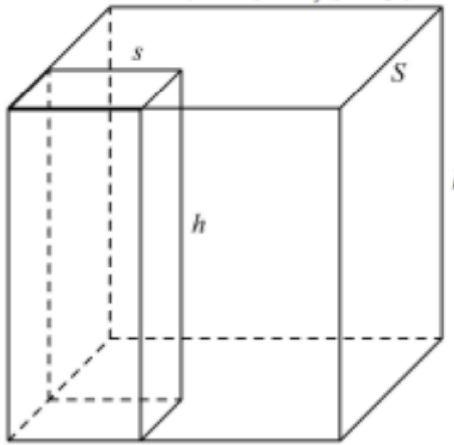
Two identical rectangular prisms each have a height of 90 centimeters. The base of each prism is a square, and the surface area of each prism is K cm².

If the two prisms are glued together along one of their square bases, the resulting solid has a surface area of $\frac{92}{47}K$ cm².

What is the side length, in centimeters, of each square base?

- _____ 8.
- | | |
|------|-------|
| a. 4 | c. 9 |
| b. 8 | d. 16 |

The formula for the difference in volumes between two square prisms of equal height is $V = hS^2 - hs^2$, where h is the height of the prisms, S is the base length of the larger prism, and s is the base length of the smaller prism. Write the formula for the difference in volumes between two square prisms in factored form. Then, find the difference in volumes with $h = 10$ cm, $S = 30$ cm, and $s = 20$ cm.



9.

- a. $V = (hS - hs)^2$; $V = 100$ c. $V = h(S^2 - s^2)$; $V = 100$
 b. $V = h(S - s)^2$; $V = 1000$ d. $V = h(S^2 - s^2)$; $V = 5000$

10.

A right rectangular prism has a square base and a volume of 2,448 cubic centimeters. If the area of the square base is 144 square centimeters, what is the area, in square centimeters, of one of the prism's four lateral faces?

- a. 17 c. 540
 b. 204 d. 816

11. A cylinder has a diameter of 8 inches and a height of 12 inches. What is the volume of the cylinder?

- a. 16π c. 192π
 b. 9π d. 768π

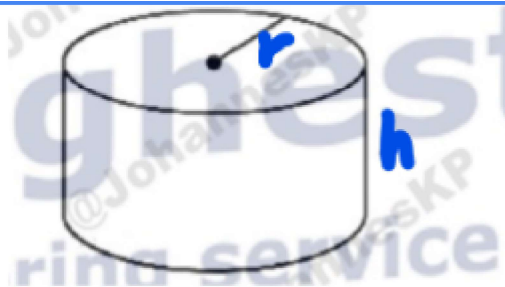
12. A cylinder has a diameter of 6 inches and a height of 24 inches. What is the volume, in cubic inches, of the cylinder?

- a. 9π c. 216π
 b. 144π d. 864π

____ 13.

A right circular cylinder has a height of 4 meters and a base with a radius of 18 meters. What is the volume, in cubic meters (m^3), of the cylinder?

- | | |
|------------|---------------|
| a. 4π | c. 288π |
| b. 22π | d. $1,296\pi$ |



The figure shows a right circular cylinder with radius r and height h . A second right circular cylinder (not shown) has a volume that is 576 times greater than the volume of the first cylinder. Which of the following expressions could represent the radius R in terms of r , and the height H in terms of h , for the second cylinder?

____ 14.

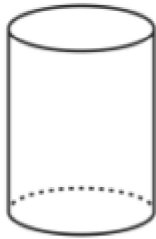
- | | |
|-----------------------|-----------------------|
| a. $R=9r$ and $H=8h$ | c. $R=8r$ and $H=9h$ |
| b. $R=9r$ and $H=64h$ | d. $R=64r$ and $H=9h$ |

A right circular cylinder has a height that is 6 inches greater than its radius. If the surface area of the cylinder is $1,080\pi$ square inches, what is the radius of the cylinder, in inches?

____ 15.

- | | |
|-------|-------|
| a. 15 | c. 21 |
| b. 18 | d. 24 |

____ 16.



A right circular cylinder has a radius r and height h .

A second cylinder has a volume **324 times greater** than that of the first.

Which expressions represent the radius R and height H of the second cylinder in terms of r and h ?

- | | |
|---------------------------|---------------------------|
| a. $R = 9r$ and $H = 6h$ | c. $R = 6r$ and $H = 9h$ |
| b. $R = 9r$ and $H = 36h$ | d. $R = 36r$ and $H = 9h$ |

____ 17.

A right square pyramid has a total surface area of **36,864 square inches**, and the combined surface area of its four lateral faces is **20,480 square inches**.

What is the **height**, in inches, of this pyramid?

- | | |
|-------|--------|
| a. 48 | c. 80 |
| b. 64 | d. 128 |

____ 18.

A right square pyramid has a total surface area of **18,816 square inches**. The lateral surface area (the combined area of the four triangular sides) is **9,600 square inches**. What is the height of the pyramid, in inches?

- | | |
|-------|-------|
| a. 14 | c. 50 |
| b. 48 | d. 96 |

____ 19. Sphere A and right circular cylinder C both have radius r . The height of cylinder C is 12 units. For what value of r will the volume of sphere A be twice the volume of cylinder C?

- | | |
|-------|-------|
| a. 18 | c. 20 |
| b. 19 | d. 21 |

____ 20. A hemisphere with radius 29 inches. Find the volume.

- | | |
|----------|----------|
| a. 16259 | c. 16261 |
| b. 16260 | d. 16262 |

Name: _____

ID: A

____ 21.

Text: "The volume of sphere A and B shown above. Find the surface area of sphere B."

Table:

Sphere	Volume
A	164π
B	20034π

- a. 7666.6
- b. 7555.5
- c. 7444.4
- d. 7333.3

____ 22. The hemisphere is half a sphere with a radius of 88 cm. Find the radius of the sphere.

- a. 88
- b. 77
- c. 66
- d. 55

____ 23. Sphere with radius 70. Find the volume of the hemisphere

- a. 228666 PI
- b. 3344777 PI
- c. 236888
- d. 234123

A hemisphere is half of a sphere.

____ 24. If the radius of a hemisphere is 83 inches, what is the closest estimate of its volume, in cubic inches?

- a. 14,400
- b. 57,700
- c. 898,200
- d. 1,197,500

____ 25.

A hemisphere is half of a sphere. If a hemisphere has a radius of 70 inches, what is the approximate volume of the hemisphere in cubic inches?

- a. 10,300
- b. 41,100
- c. 538,800
- d. 718,400

Name: _____

ID: A

The formula above shows the relationship between the volume V of a pyramid, the area A of its base, and its height h :

$$V = \frac{1}{3}Ah$$

If this equation is rewritten in the form

$$h = k \cdot \frac{V}{A}$$

where k is a constant, what is the value of k ?

____ 30.

- a. 3
- b. 4

- c. 5
- d. 6

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Answer Section****MULTIPLE CHOICE**

- | | |
|------------|--------|
| 1. ANS: D | PTS: 1 |
| 2. ANS: D | PTS: 1 |
| 3. ANS: A | PTS: 1 |
| 4. ANS: D | PTS: 1 |
| 5. ANS: A | PTS: 1 |
| 6. ANS: D | PTS: 1 |
| 7. ANS: D | PTS: 1 |
| 8. ANS: B | PTS: 1 |
| 9. ANS: D | PTS: 1 |
| 10. ANS: B | PTS: 1 |
| 11. ANS: C | PTS: 1 |
| 12. ANS: C | PTS: 1 |
| 13. ANS: D | PTS: 1 |
| 14. ANS: C | PTS: 1 |
| 15. ANS: A | PTS: 1 |
| 16. ANS: C | PTS: 1 |
| 17. ANS: A | PTS: 1 |
| 18. ANS: A | PTS: 1 |
| 19. ANS: A | PTS: 1 |
| 20. ANS: A | PTS: 1 |
| 21. ANS: A | PTS: 1 |
| 22. ANS: A | PTS: 1 |
| 23. ANS: A | PTS: 1 |
| 24. ANS: D | PTS: 1 |
| 25. ANS: D | PTS: 1 |
| 26. ANS: D | PTS: 1 |
| 27. ANS: A | PTS: 1 |
| 28. ANS: A | PTS: 1 |
| 29. ANS: A | PTS: 1 |
| 30. ANS: A | PTS: 1 |